

BAKHMUT, Ukraine

WMO: 345100

Lat: 48.5572N

Lon: 38.0147E

Elev: 407

StdP: 14.48

Time Zone: 2.00 (EUE)

Period: 84-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.8	(c) 0.1	(d) -10.4	(e) 3.2	(f) -5.0	(g) -4.7	(h) 4.3	(i) 0.6	(j) 15.6	(k) 32.8	(l) 13.8	(m) 29.1	(n) 2.4	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.5	(c) 90.3	(d) 68.9	(e) 87.1	(f) 67.3	(g) 84.0	(h) 66.0	(i) 70.7	(j) 85.0	(k) 69.1	(l) 83.1	(m) 67.7	(n) 80.7	(o) 5.1	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	98.5	75.7	64.6	92.7	73.9	63.1	87.8	72.7	34.8	85.0	33.5	82.9	32.3	80.7	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 14.5	(b) 12.0	(c) 10.2	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.8	25.1	26.0	35.0	49.3	61.7	66.8	70.0	68.9	59.2	47.1	35.3	27.4
	DBStd	18.78	11.13	12.30	9.10	6.95	6.20	5.58	5.15	5.84	7.70	9.20	10.58	10.04
	HDD50	3333	772	674	465	97	2	0	0	0	12	165	446	700
	HDD65	6770	1237	1093	929	471	141	45	13	26	206	554	890	1165
	CDD50	2524	0	1	2	76	364	505	620	586	287	76	6	1
	CDD65	483	0	0	0	0	38	100	168	146	31	0	0	0
	CDH74	5378	0	0	0	9	381	913	1847	1783	437	8	0	0
	CDH80	1794	0	0	0	0	55	254	662	692	131	0	0	0
(14) Wind	WSAvg	4.2	4.7	4.4	5.2	5.1	4.0	3.8	3.6	3.3	3.6	4.0	4.1	4.6
(15) Precipitation	PrecAvg	31.5	3.6	3.0	2.8	2.1	2.4	2.9	2.6	1.8	2.0	2.0	2.7	3.7
	PrecMax	42.8	6.9	7.2	5.7	4.8	5.4	5.7	6.1	3.4	5.7	4.3	5.3	7.0
	PrecMin	24.7	1.3	0.7	1.3	0.4	0.3	0.6	0.4	0.3	0.2	0.7	0.3	1.5
	PrecStd	5.0	1.5	1.3	1.1	1.0	1.1	1.3	1.3	0.9	1.3	0.8	1.3	1.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.7	56.3	63.3	75.2	84.7	89.1	94.3	95.3	92.5	75.8	59.9	52.8
		MCWB	42.0	49.2	49.6	57.9	64.2	68.0	69.9	69.7	68.3	59.6	54.0	51.7
	2%	DB	41.6	47.5	56.4	69.7	81.6	86.0	89.9	90.8	82.6	70.9	55.6	44.8
		MCWB	40.1	44.2	46.3	55.7	62.4	67.1	69.9	68.3	63.2	57.5	52.1	42.9
	5%	DB	39.5	42.4	50.9	66.4	78.9	83.1	86.6	87.3	78.7	65.8	51.6	41.5
		MCWB	38.0	39.8	44.4	52.8	62.3	67.2	68.3	66.3	62.3	55.3	48.6	39.9
	10%	DB	36.8	39.1	46.9	62.8	75.3	79.4	83.1	83.8	74.5	60.9	48.0	37.8
		MCWB	35.4	37.2	41.9	51.1	61.2	65.7	66.3	65.5	60.2	54.1	45.1	36.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.8	49.2	50.4	59.3	67.3	72.6	74.1	71.2	69.0	61.7	54.8	51.9
		MCDB	43.2	53.6	59.6	72.2	78.0	82.8	87.1	88.0	87.1	72.2	58.1	52.8
	2%	WB	40.4	44.4	47.3	56.6	65.8	70.0	71.4	69.6	66.1	59.5	52.2	43.4
		MCDB	41.3	46.7	54.4	67.9	76.7	81.2	86.4	85.7	79.9	66.5	55.8	44.5
	5%	WB	38.1	39.9	45.1	54.7	64.2	68.3	69.3	68.2	63.1	57.1	49.0	40.1
		MCDB	39.5	41.8	50.8	63.2	74.3	79.6	83.7	83.3	74.6	63.4	51.0	41.6
	10%	WB	35.5	37.3	42.2	52.3	62.5	66.7	67.6	66.8	61.1	54.6	45.3	36.6
		MCDB	36.7	39.4	46.3	60.2	72.8	77.0	80.1	80.7	72.2	60.6	47.1	37.7
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	12.4	13.6	18.7	21.3	20.3	21.5	23.9	22.5	18.4	10.9	8.8
		MCDBR	9.2	15.2	21.8	24.9	25.6	24.8	26.8	29.8	30.6	26.0	14.5	9.0
		MCWBR	8.1	11.8	13.7	13.0	11.5	10.5	10.7	11.8	13.9	14.5	9.7	8.1
	5% WB	MCDBR	8.3	13.0	19.8	21.0	21.0	20.9	24.6	26.0	26.1	21.2	12.6	8.7
		MCWBR	7.8	10.4	13.1	12.2	10.1	10.1	10.9	11.1	13.4	13.1	9.5	8.4
(40) Clear-Sky Solar Irradiance	taub		0.331	0.362	0.395	0.460	0.421	0.418	0.445	0.443	0.414	0.378	0.354	0.334
	taud		2.170	2.099	2.142	2.047	2.244	2.283	2.216	2.206	2.250	2.315	2.288	2.225
	Ebn at Noon		221	241	255	250	267	268	259	253	247	236	213	203
	Edh at Noon		26	36	40	49	42	41	43	42	36	29	24	22
(44) All-Sky Solar Radiation	RadAvg		309	607	946	1376	1839	1976	1944	1762	1224	706	324	200
	RadStd		66	125	142	144	158	170	126	123	129	109	62	30

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	+1.26	N/A	N/A	N/A	+1.96	N/A	N/A	N/A	-324	+218	+142

Nomenclature: See separate page